Lecture 19: Introduction to Planar Graphs

Date: April 6, 2026 **Scribe:** Jon Terry

This lecture introduces the main definitions and concepts related to planar graphs.

1 Motivation

We can motivate our investigation of planar graphs by asking whether K_5 (the complete graph on five nodes) can be drawn in the plane without edge crossings.

Notice that when we say that two edges "cross", we mean a situation like this:

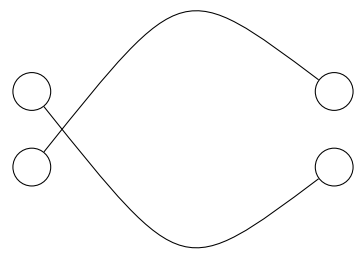


Figure 1: Two edges crossing.

We will use proof-by-picture to answer this question. First, consider the cycle obtained by connecting the five nodes of K_5.

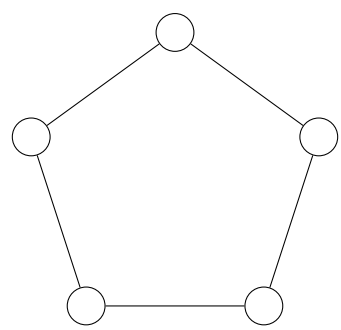


Figure 2: The five-cycle C_5 from K_5.

Now, the five remaining edges in K_5 must each appear either inside or outside the boundary curve we just defined. Let's start with the interior. Add any interior edge.

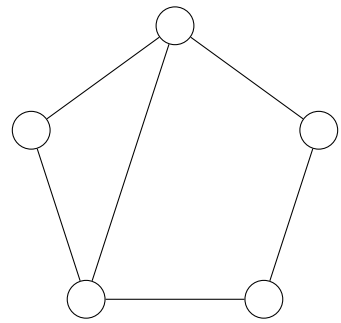


Figure 3: C_5 with one interior edge.

We see from the picture that given one interior edge, we can add at most one more interior edge without creating a crossing. Thus, we can add at most two interior edges. Similarly, we can add up to two exterior edges without creating a crossing.

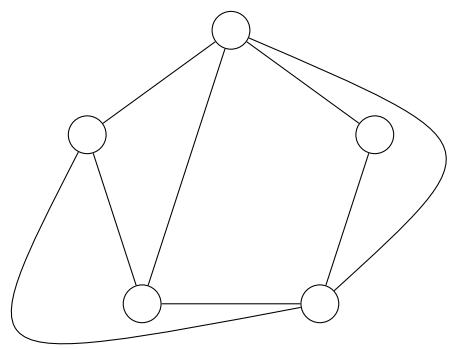


Figure 4: C_5 with one interior edge and two exterior edges.

Thus, based on this particular example, it appears that a drawing of a graph on five nodes can contain up to $5 + 2 + 2 = 9$ edges without creating a crossing. But K_5 has $\binom{5}{2} = 10$ edges. Therefore, at least provisionally, the answer to the question "can you draw K_5 in the plane without edge crossings?" seems to be "no".

2 Preliminary Definitions

Let's establish some terminology to put this discussion on a more rigorous foundation.

Definition 1 (Planar curve)**.** A *curve* in the plane is the image of a continuous function $f : [0, 1] \to \mathbb{R}^2$. We can adopt this as our definition of how we draw an edge of a graph in the plane, since it gives us the ability to connect nodes using curves.

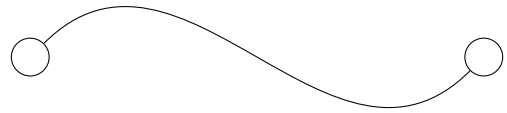


Figure 5: A planar curve in $\mathbb{R}^2$ connecting two nodes.

Definition 2 (Closed curve)**.** A curve is closed if it starts and ends at the same point, i.e., $f(0) = f(1)$.

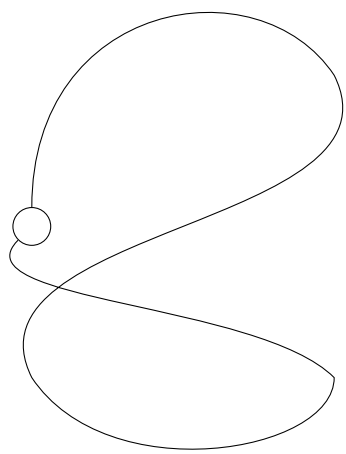
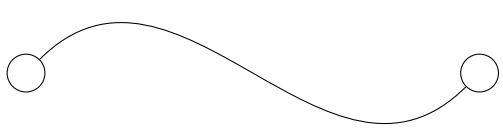
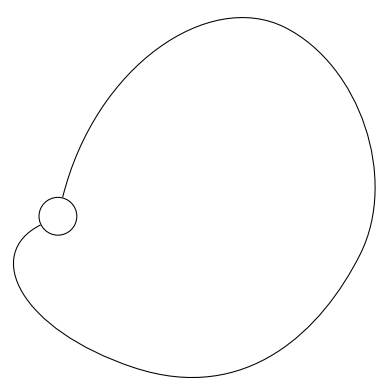


Figure 6: A closed curve.

Definition 3 (Simple curve)**.** A curve is *simple* if the only point it revisits is its starting point.



(a) A simple curve.



(b) A simple closed curve.

Definition 4 (Edge crossing)**.** A *crossing* of two curves occurs when the two curves share points in common other than their endpoints. We showed an example of an edge crossing in Figure 1.

Importantly, we see from these definitions that a closed curve can intersect itself, while, on the other hand, a simple curve can be closed but cannot intersect itself.

Definition 5 (Graph drawing)**.** A drawing of a multigraph $G = (V, E)$ maps V to points in $\mathbb{R}^2$ and E to simple curves that preserves the incidence relations of G. From here forward, when these notes refer to a graph G, assume that G is a multigraph.

Definition 6 (Graph embedding)**.** An *embedding* of a graph in a surface S is a mapping of vertices to points in S and edges to simple curves in S (i.e., without crossings) that preserves incidence.

Thus, an embedding in $\mathbb{R}^2$ is a drawing without crossings. Unless otherwise stated, "embedding" means an embedding in $\mathbb{R}^2$. We can now rephrase the motivating question as: "Does K_5 have an embedding?"

3 Planar Graphs

With these definitions in place, we can now state the following.

Definition 7 (Planar graph)**.** A graph $G = (V, E)$ is *planar* if it has an embedding in $\mathbb{R}^2$.

Definition 8 (Plane graph)**.** A *plane graph* is any particular embedding of a planar graph in $\mathbb{R}^2$. In other words, a plane graph refers to an *embedding* of G in the plane, not G itself.

At this point, we need to add two topological definitions to complete the description of the components of plane graphs.

Definition 9 (Open set)**.** A set $U \subset \mathbb{R}^2$ is *open* in $\mathbb{R}^2$ if, for any point $p \in U$, there is an $r > 0$ such that for all points $x \in \mathbb{R}^2$ within distance r of p, those points are also in U.

Definition 10 (Region)**.** A *region* is an open set $U \subsetneq \mathbb{R}^2$ such that every two points in U lie on a curve contained in U.

Definition 11 (Faces of a plane graph)**.** The *faces* of a plane graph are the regions of the plane obtained by removing G. In other words, the faces of a plane graph are the regions of $\mathbb{R}^2 \setminus G$.

As we proceed with our discussion of planar graphs, it will become important to be able to distinguish between the interior and exterior faces of a plane graph. We appeal to the Jordan curve theorem (which we state without proof) to make this distinction.

Theorem 3.1 (Jordan curve theorem)**.** *Let C be a simple, closed curve and consider the set $A = \mathbb{R}^2 \setminus C$. This set A consists of two regions. One region is bounded, which we call the* interior*, and one region that is unbounded, which we call the* exterior*. Both of these regions share the same boundary, C.*

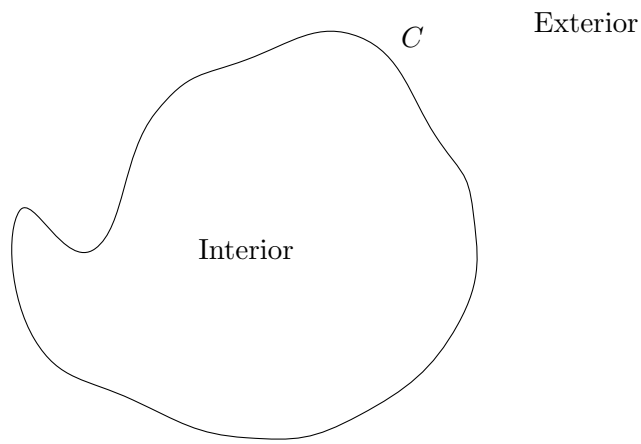
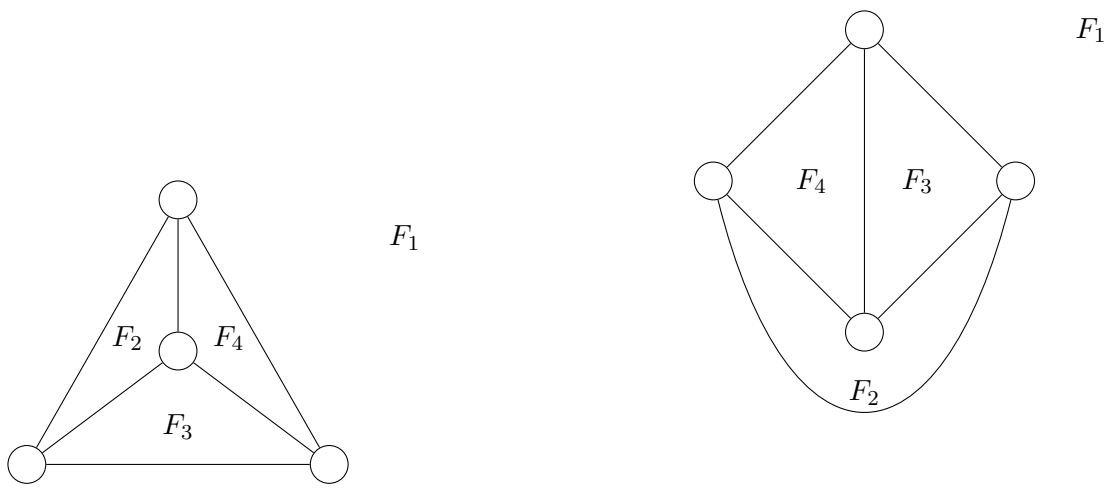


Figure 8: The Jordan curve theorem: a simple closed curve C divides $\mathbb{R}^2$ into an interior and an exterior.

Example 1. The complete graph K_4 is a planar graph. Below are two plane embeddings of K_4 with their faces numbered. Note that in the case of finite graphs (which are the only graphs we will consider here), every plane embedding induces an unbounded face, which we have labeled F_1 in both drawings.



(a) Embedding A

(b) Embedding B

Figure 9: Two plane embeddings of K_4 with faces labeled.

We can make one further distinction among planar graphs and their embeddings.

Definition 12. A planar graph $G = (V, E)$ is *outerplanar* if it has an embedding such that every $v \in V$ is on the boundary of the unbounded face.

Finally, since we are only concerned with finite graphs, we can restrict ourselves to curves C in the plane that consist of the union of finitely many line segments. This leads to the next definition.

Definition 13 (Polygonal curve)**.** A *polygonal* curve is a curve in the plane that consists of the union of finitely many line segments.

When working with polygonal curves, we can use a construction in the plane to allow us to distinguish between the interior and exterior regions of the plane induced by a planar graph. The construction works like this. Pick a point $x \in \mathbb{R}^2$. Draw a ray from x that passes through the curve but does not intersect any of its segment endpoints. If the parity of the number of intersections of the ray with the curve is even, then the point is in the exterior region defined by the curve. If the parity of the number of intersections is odd, then the point is in the interior region defined by the curve.

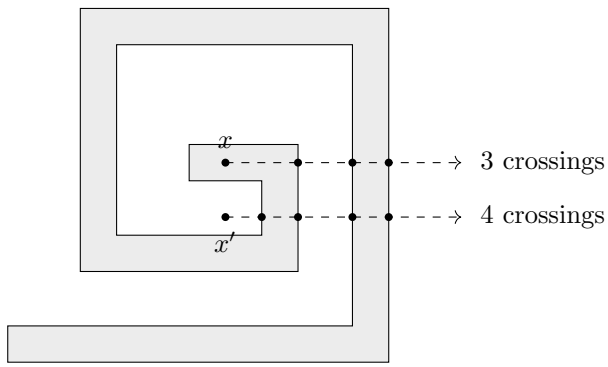


Figure 10: The ray-casting parity test. A ray from x (inside the worm) crosses the boundary an odd number of times. A ray from x' (outside the worm) crosses the boundary an even number of times.

4 Dual of a plane graph

Every plane graph gives rise to another plane graph, which we call the *dual* of the original graph.

Definition 14 (Dual of plane graph)**.** The *dual* of a plane graph G is another plane graph G^* whose nodes are the faces of G and whose edges correspond to the edges of G such that two nodes in G^* are adjacent if and only if the corresponding faces in G share a boundary edge.

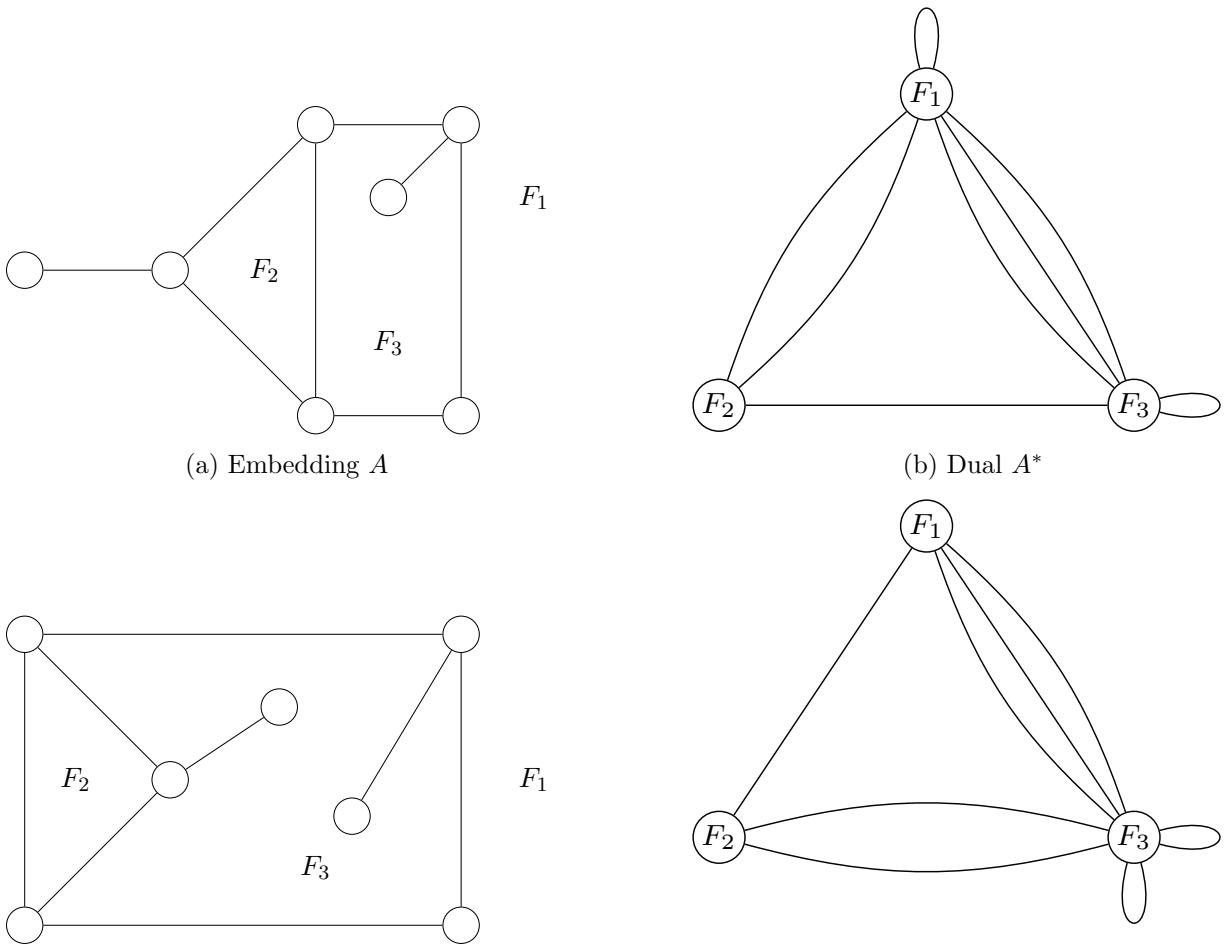
We call G^* the dual of G, since $(G^*)^* \simeq G$.[1] However, the duals of different plane embeddings of the same planar graph need not be isomorphic to each other. The following definition provides a way to see this.

Definition 15 (Length of a face)**.** The length of a face in a plane graph G is the total number of steps of the closed walk in G that bounds the face. If F is a face of a plane graph, then we write $\ell(F)$ for the length of face F.

Example 2. Consider two different plane embeddings of a planar graph as shown in the figure below. We see that the duals of the plane embeddings A and B are not isomorphic. To see why, we can just compare the lengths of the corresponding faces in both duals:

	$\ell(F_1)$	$\ell(F_2)$	$\ell(F_3)$	Total
A^*	7	3	6	16
B^*	4	3	9	16

[1]Recall from lecture 1 that two graphs $G = (V(G), E(G))$ and $H = (V(H), E(H))$ are isomorphic if there exists a bijection $\sigma : V(G) \to V(H)$ such that $(u, v) \in E(G)$ if and only if $(\sigma(u), \sigma(v)) \in E(H)$.



(a) Embedding A (b) Dual A^*

(c) Embedding B (d) Dual B^*

Figure 11: Two plane embeddings of a planar graph with their duals.

The fact that the sums of the lengths of the faces of both duals are equal is not a coincidence. We conclude the lecture with the following proposition.

Proposition 4.1. *In a plane graph $G = (V, E)$ with $|E| = m$, the sum of the lengths of the faces is twice the number of edges in G. More formally, let n be the number of faces in G and let $|E| = m$. Then $\sum_{i=1}^{n} \ell(F_i) = 2m$.*

Proof. Recall the handshaking lemma, which states that for any graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|,$$

since every edge contributes 2 to the total degree count (one for each endpoint). Next, note that $|E(G)| = |E(G^*)| = m$, and that $\ell(F_i) = \deg_{G^*}(F_i)$ for all faces i in G. Combining these three facts into one identity yields the desired result:

$$\sum_{i} \ell(F_i) = \sum_{F_i \in V(G^*)} \deg_{G^*}(F_i) = 2m.$$

□